Code No: 136AG **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, March - 2024
ARTIFICIAL INTELLIGENCE
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What are the four categories under which AI is classified?	[2]
b)	What do you mean by Game Playing?	[3]
c)	Define well-formed formula.	[2]
d)	What is frames?	[3]
e)	Write about Expert systems.	[2]
f)	Why are heuristics crucial for the efficient design of an expert system?	[3]
g)	What are the basic design issues and approaches to machine learning?	[2]
h)	What is the role of radial basis function in separating nonlinear patterns?	[3]
i)	What is Natural language processing?	[2]
j)	Mention the types of Parsers.	[3]

PART – B

(50 Marks)

2. Explain in the following detail with examples:
a) Iterative deepening search
b) Heuristic search. [5+5]

OR

3. How to define a problem as state space search? Discuss it with the help of an example. [10]

4. Describe Knowledge Representation using Semantic Network in detail. [10]

OR

5. Consider the following facts
a) Team India
b) Team Australia
c) Final match between India and Australia
d) India scored 350 runs, Australia scored 350 runs, India lost 5 wickets, Australia lost 7 wickets.
e) The team which scored the maximum runs wins.
f) If the scores are same the team which lost minimum wickets wins the match.
Represent the facts in predicate, convert to clause form and prove by resolution "India wins the match". [10]

6. Write a short note on:
   a) Justification – Based Truth Maintenance Systems
   b) Logic – Based Truth Maintenance. [5+5]

OR

7. Explain the phases in Building Expert Systems in detail. [10]

8.a) Give decision trees to represent the following boolean functions:
   i) A ∧ ˜B
   ii) A ∨ [B ∧ C]
   iii) A XOR B
   iv) [A ∧ B] ∨ [C ∧ D]

   b) Explain perceptron learning algorithm. [5+5]

OR

9.a) Compare K means clustering with Hierarchical Clustering Techniques.

   b) Differentiate between Supervised, Unsupervised and Reinforcement Learning. [5+5]

10. Explain how the natural language is processed, with a relevant example. [10]

OR

11. Describe the steps involved in the knowledge engineering process with example. [10]

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